

Opponent Modeling

One word, nine senses, and the nine places two passes disagreed

Your Headspace page asks the question this file exists to make answerable.

“How exactly can opponent modeling benefit network defense? What network security (or specifically, adversary emulation, threat modeling, etc) constrains and demands pose interesting challenges on opponent modeling?”

what-is-she-thinking.md, added 2026-08-26

Your dictionary page already puts opponent modeling, threat modeling and adversary emulation side by side, as three siblings placed on five dimensions. The Headspace question puts two of them underneath the third, as the source of its constraints. Both placements are yours. They are days apart, and they are not the same claim.

This file keeps the table and adds to it. Section 2 names the nine senses the word carries across your files and across the three fields that use it. Section 3 lists forty occurrences in your own writing and nine anchors from the fields, and assigns each one a sense. Section 4 assigns them all a second time, with the file names taken away. Section 5 is the part worth reading first. It is the nine occurrences where the two assignments disagreed, and they stay disagreeing. Which sense you meant is your call, not the census’s.

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1 Your table

Reproduced without change from the Cyber-Human-AI Dictionary page. Everything after this section is built on it.

Concept	What is represented?	Why represent it?	Relationship to reality	Output
Opponent modeling	decision process, strategy, policy, beliefs/type	predict/respond strategically	infer/learn a model of an interacting agent	model used for decision-making
Threat modeling	threats, adversaries, capabilities, attack paths, assets/exposure	identify and reason about possible harm	characterize a threat space, often prospectively	structured account of threats/risk
Adversary emulation	observable behaviors/TTPs of an adversary	test defenses against realistic behavior	deliberately reproduce selected real-world behavior	enacted adversarial behavior

And the five dimensions the page places them on.

1. **Object.** actor, behavior, attack possibilities, environment.
2. **Epistemic operation.** describe, infer, predict, simulate/enact.
3. **Specificity.** generic adversary, adversary class, particular actor.
4. **Purpose.** strategic response, risk analysis, defensive evaluation.
5. **Fidelity.** abstract representation, executable reproduction.

The five dimensions are what make the census possible. Every sense below is placed by naming its Object, its Epistemic operation and its Output, and two senses are the same sense when those three agree.

2 Nine senses

Four fields each, in your order. Definition. Distinguishes it from. Operational test. In my project.

S1. Learned opponent representation

Definition. A vector inferred from the modeling agent’s own observation stream, trained to predict something about the other agent, and consumed by a policy.

Distinguishes it from. S2. There is no named set of opponents. The representation is whatever the loss makes it, and it carries no guarantee of being interpretable as a type.

Operational test. Is there a decoder or a classifier head that is used in training and thrown away before deployment? If yes, S1.

In my project. FOE-Dreamer’s τ . The per-device $z_t^{\text{opp},i}$ of PickYourBattles. The opponent latent on the Cyber World Modeling pages.

S2. Type inference

Definition. A posterior over a set of opponent types that was fixed before play began, updated from observed behavior.

Distinguishes it from. S1. The set exists first and the inference returns a distribution over it. From S5, because the set is absolute and does not shift when the population around the opponent changes.

Operational test. Can you write down the types before the first episode, and does the model return a number per type? If yes, S2.

In my project. The word “type” in the Approach section of the FOE-Dreamer paper, and in `next.md`’s “recovering a hidden type from behaviour”.

S3. Best response to a fixed strategy

Definition. The opponent is a strategy already sitting in a population, and modeling it means computing a best response to it and putting the answer back in the population.

Distinguishes it from. S1 and S2. Nothing is inferred. The opponent is handed to you, and the operation is optimization rather than estimation.

Operational test. Does the output of the modeling step go into a strategy pool rather than into a policy’s input? If yes, S3.

In my project. AcceleratePSRO, the whole loop. Learn Structure’s opponent-blindness test, which is the decision that this operation can be skipped for a region.

S4. Behavioral adversary model

Definition. A parameterized departure from expected-utility maximization, fitted to or imposed on the choices of a human adversary.

Distinguishes it from. S2. The parameters have psychological names and vary continuously. A type is a label; a behavioral model is a curve.

Operational test. Does the model have a parameter you could report to a psychologist, such as a loss-aversion multiplier or a response precision? If yes, S4.

In my project. The Biased Attacker page, in full. Prospect theory’s value function inside the POMDP, with α , β and λ .

S5. Categorization by a bounded observer

Definition. A small number of categories assigned by an agent that will not carry one model per partner, with the assignment made relative to the group rather than against a fixed threshold.

Distinguishes it from. S2. The categories move when the population moves. The same behavior lands in a different category depending on who else is in the room.

Operational test. Hold a partner’s behavior fixed and change the group around them. If the category changes, S5. If it does not, S2.

In my project. The Iterated Prisoner’s Dilemma page. Cooperative, conditional, exploitative, assigned against the group mean. ReadTheRoom’s strategy detection question.

S6. Theory of mind

Definition. Attribution of unobservable mental states to the other agent, including that agent's beliefs about you, to a stated depth.

Distinguishes it from. S1 through S5. The model contains a model of the modeler. Depth is a property of the representation and has to be declared.

Operational test. Does the representation include the opponent's model of your policy? If yes, S6.

In my project. Not used. It is the sense the cognitive science literature reaches for first, and naming it here is what keeps it from being read into the other eight.

S7. Threat model

Definition. A prospective structured record of protected assets, system boundaries, adversary goals, knowledge and capabilities, reachable attack surface, adverse events, likelihood assumptions, and controls.

Distinguishes it from. S8. It is written before the exercise and it states what is out of scope. It is a document, not a run.

Operational test. Does it contain a list of things it declines to consider? If yes, S7.

In my project. Section 4 of the FOE-Dreamer paper, and the vocabulary table in `threat_modeling_red_teaming`.

S8. Adversary emulation

Definition. Reproduction of a documented actor's tactics, techniques and procedures so that a red team can test detection and response against that actor's behavior.

Distinguishes it from. S7, which characterizes a space of threats rather than enacting one. From red teaming, which searches for any way to defeat the system; emulation is the narrower case that fixes one adversary's known behavior.

Operational test. Can you name the real actor being reproduced and cite the report it came from? If yes, S8.

In my project. Not used as a method. Named twice: on the dictionary page as a sibling concept, and in the Headspace question as a source of constraints.

S9. Attacker construction

Definition. Writing the opponent you will train or evaluate against, with its behavior specified by you and its ground truth available to you by construction.

Distinguishes it from. S8. The fidelity target is a behavioral property, sophistication, bias, patience, rather than a named actor. There is no report to cite because the actor does not exist.

Operational test. Did you write the attacker, and can you read its internal state out of the simulator? If yes, S9.

In my project. B-line and Meander. Avoidant and Enticed. The three prospect-theory profiles. This is the sense your table has no row for, and it is the one that appears most often in your own files.

3 The census

Forty occurrences in your files, then nine anchors from the three fields. The sense column is the first assignment.

3.1 Your files

	The occurrence	Where	Sense
O1	how exactly can opponent modeling benefit network defense	Headspace	S1
O2	network security, or specifically, adversary emulation, threat modeling, as the source of constraints and demands	Headspace	S7, S8
O3	approaches that help differentiate human vs AI adversaries	Headspace	S2
O4	infer/learn a model of an interacting agent	Dictionary, table row 1	S1
O5	characterize a threat space, often prospectively	Dictionary, table row 2	S7
O6	deliberately reproduce selected real-world behavior	Dictionary, table row 3	S8
O7	how agents read each other into kinds across repeated interaction	Mental World Modeling README	S5
O8	an attacker whose biases are built into the reward	Mental World Modeling README	S9
O9	recovering a hidden type from behaviour, while the other party adapts	<code>next.md</code>	S2
O10	one account of how an agent types another, across cooperative and adversarial play	<code>next.md</code>	S5
O11	it compresses them into a small number of types	Iterated Prisoner's Dilemma	S5
O12	the categories are relative, not absolute	Iterated Prisoner's Dilemma	S5
O13	nobody sees the strategy list, they only see what happened last round	Iterated Prisoner's Dilemma	S1
O14	real attackers are not expected-utility maximisers, most models of them are	Biased Attacker	S4
O15	put the bias in the reward function	Biased Attacker	S4
O16	behavioural fingerprints, you could tell one kind of adversary from another by how they move	Biased Attacker	S2
O17	train defenders against a population rather than a single idealised opponent	Biased Attacker	S9
O18	a fixed route, a prepared sequence of actions, with a jump table	Challenging Attacker, B-line	S9
O19	breadth first, it scans every subnet it can see	Challenging Attacker, Meander	S9
O20	an opponent embedding inferred from defender-observable signals alone	<code>main.tex</code> , abstract	S1
O21	the opponent embedding τ encodes attacker type and recent attacker behavior	<code>main.tex</code> , Approach	S1
O22	attacker behavior leaves signatures in the defender's telemetry	<code>main.tex</code> , Opponent Model	S1
O23	lets the policy condition on inferred type rather than wait for the attacker to commit a revealing action	<code>main.tex</code> , Opponent Model	S2
O24	supervised against attacker class via a small classifier head on τ_t	<code>main.tex</code> , Opponent Model	S2
O25	we model two fixed-class profiles that bracket the spectrum of attacker sophistication	<code>main.tex</code> , Threat Model	S9
O26	several threats are out of scope, we do not consider adaptive learning adversaries	<code>main.tex</code> , Threat Model	S7
O27	the opponent model stays separable and inspectable	FOE-Dreamer website page	S1

	The occurrence	Where	Sense
O28	a compact representation of who the agent is up against	Cyber World Modeling README	S1
O29	whether drift in the opponent latent can be detected before it costs	Cyber World Modeling <code>next.md</code>	S1
O30	imagined rollouts can stand in for expensive environment interaction when computing a best response inside PSRO	AcceleratePSRO	S3
O31	the opponent can be folded into the environment as one more stochastic process	<code>taxonomy.tex</code>	none
O32	opponent-blindness, as one of the four conditions that excuse a structure	<code>taxonomy.tex</code>	S3
O33	none of the four is defined for an adapting adversary	<code>taxonomy.tex</code>	S3
O34	control dependence fixes which of the opponent’s inputs must be answered	<code>strategic_dependence.tex</code>	S3
O35	per-device opponent embedding, inferred attacker behavior at device i	PickYourBattles §6.3.3	S1
O36	can agents categorize and adapt to different opponent strategies	ReadTheRoom README	S5
O37	agents learn averaged responses that perform moderately against all opponent types but excel against none	ReadTheRoom report	S2
O38	an explicit record of protected assets, system boundaries, adversary goals, knowledge and capabilities	<code>threat_modeling_red_teaming.tex</code>	S7
O39	the emulation fixes an adversary profile and behavior chain rather than searching arbitrarily for any failure	same file	S8
O40	red teaming searches for ways to defeat the system, adversary emulation is the narrower case	same file	S8

Two counts worth having before the second pass. S1 and S9 together take twenty-one of the forty. S6 takes none. The sense the cognitive science shelf would reach for first is the one your corpus never uses, and the sense your dictionary table has no row for is the one it uses most.

3.2 The three fields

Nine anchors, one per lineage the word travels in. These are what the senses above are pinned to, so that a reader from any of the three can find their own usage.

Multi-agent AI “*constructing models which make predictions about various properties of interest, such as actions, goals, beliefs, of the modelled agents*”

L1. Albrecht and Stone, Autonomous agents modelling other agents, *Artificial Intelligence* 258, 66 to 95, 2018. The survey sorts the methods into policy reconstruction (§4.1), type-based reasoning (§4.2), classification (§4.3), plan recognition (§4.4), recursive reasoning (§4.5), graphical models (§4.6) and group modelling (§4.7). S1 is §4.1, S2 is §4.2, S5 is §4.3, S6 is §4.5.

Deep RL “*the encoder reads the controlled agent’s local observations alone, the decoder that reconstructs the modelled agent’s trajectory is used in training and dropped at execution*”

L2. Papoudakis, Christianos and Albrecht, Agent Modelling under Partial Observability for Deep Reinforcement Learning, NeurIPS 2021. The architectural parent of S1, and the source of the terms *controlled agent*, *modeled agent* and *local information*.

Deep RL, against S1 *“Most opponent modeling methods deal with the non-stationarity caused by unknown opponent policies via predicting the opponent’s actions. However, focusing on the opponent’s action is shortsighted, which also constrains the adaptability to unknown opponents in complex tasks.”*

L3. Yu, Jiang and Lu, Opponent Modeling based on Subgoal Inference, NeurIPS 2024, abstract.

Game theory *“a player’s private information is a type, each player knows their own and holds a distribution over the others’, and payoffs depend on the whole type profile”*

L4. Harsanyi, Games with Incomplete Information Played by Bayesian Players, *Management Science* 14(3), 1967. The origin of S2, and of the word your Approach section borrows when it writes attacker type.

Game theory, applied *“equilibrium play is unexploitable and does not exploit, opponent modeling is the deliberate move away from it toward a best response against a specific weakness”*

L5. Billings, Papp, Schaeffer and Szafron, Opponent Modeling in Poker, AAAI 1998; Southey et al., Bayes’ Bluff, UAI 2005. This is where S3 and S2 sit on top of each other, since the best response is computed against an inferred type.

Security games *“attacker bounded rationality is modelled as a stochastic response whose probability rises with a target’s expected value, with SUQR adding a linear combination of domain features”*

L6. The quantal response and SUQR line in Stackelberg security games. The field anchor for S4, and the reason S4 is a separate sense rather than a shade of S2.

Security, prospective *“threat modeling is analyzing representations of a system to highlight concerns about security and privacy characteristics”*

L7. Threat Modeling Manifesto, 2020. Shostack’s four questions are the same definition in operational form: what are we working on, what can go wrong, what are we going to do about it, did we do a good job. The anchor for S7.

Security, enacted *“prototype documents of what can be done with publicly available threat reports and ATT&CK, which allow defenders to test their networks and defenses by enabling red teams to more actively model adversary behavior”*

L8. MITRE ATT&CK adversary emulation plans. The anchor for S8, and the reason S8’s operational test asks for a named actor.

Cyber defense, explicit belief *“an explicit Bayesian belief over attacker location and phase, with tree search run over the belief”*

L9. Causal-POMCP for CAGE-2, arXiv 2407.11070. The alternative to S1 within the same problem: an explicit distribution over attacker state instead of a learned vector, which is S2 built by hand.

4 The second pass

Every occurrence in Section 3 was assigned twice. The second assignment saw the sentence and nothing else. No file name, no neighbouring sentence, no author, and no first verdict. The two assignments were then laid side by side.

Forty of the forty-nine agree. Those forty are settled and need no further argument; the sense in the table is the sense.

Nine disagree. They are Section 5, and they are not resolved there. A disagreement between two honest assignments of the same sentence is evidence that the sentence carries two senses, not evidence that one of the assignments was careless. Every one of the nine is a place where a reader from a different field would read your sentence differently from the way a reader from your own field reads it.

5 Ambiguous

D1 O1, the headword itself

First pass S1, second pass S7 with S8.

The sentence is “How exactly can opponent modeling benefit network defense?” Read on its own, the beneficiary is network defense, and network defense’s own word for reasoning about adversaries is threat modeling. Read next to your FOE-Dreamer work, opponent modeling is τ and the answer is a policy input. The sentence that follows it in your file names adversary emulation and threat modeling explicitly, which is what pulled the second pass toward S7.

This one is the whole file in miniature. Your question cannot be answered until it says which sense it is asking about, and it currently supports two.

D2 O3, human versus AI adversaries

First pass S2, second pass S4. Neither holds.

“Are there any existing opponent modeling approaches can help differentiate human vs AI adversaries?” S2 says human and AI are two types and the model returns a posterior over them. S4 says the difference is a parameter of a behavioral model. Both are defensible and both are wrong about the object, because what is to be recovered is neither a policy nor a preference. It is the algorithmic making of the agent, and your Headspace page names that as a separate question one bullet later: does it matter whether an agent is rule-based, RL-based, or LLM-based, if their behavior are similar.

No sense in the census has algorithmic provenance as its Object, and no field anchor in Section 3.2 supplies one. This is a gap in the five dimensions, not a tenth sense. The Object axis runs actor, behavior, attack possibilities, environment, and the thing this question asks about is none of the four.

D3 O8, biases built into the reward

First pass S9, second pass S4.

“how to model an attacker whose biases are built into the reward, so that the defender it trains faces a realistic opponent instead of an idealized one.” Both passes are right. The attacker is written by you, so its ground truth is available by construction, which is S9. Its parameters are α , β and λ , which is S4. The Biased Attacker page is the one place in the corpus where a behavioral model is used as a construction method rather than as an inference target.

The ambiguity is the design, and it is worth keeping. Fitting prospect theory to a real attacker and imposing prospect theory on a scripted one are opposite operations that produce the same object.

D4 O16, behavioural fingerprints

First pass S2, second pass S5.

“you could tell one kind of adversary from another by how they move” is a claim about kinds, and kinds are types. But your own Iterated Prisoner’s Dilemma page says categories are assigned against the group mean and not against a fixed threshold. A fingerprint is absolute. A category on that page is relative. Move a mildly cooperative partner into a generous group and they become the exploiter, and no fingerprint behaves that way.

So the two pages use “kind” under two incompatible assignment rules, and the operational test in S5 separates them cleanly. Which rule the Biased Attacker page intends, the page does not say.

D5 O21 and O24, one symbol and three senses

First pass S1, second pass S2, and a third reading of S9 that neither pass returned on its own. τ is a vector produced by an encoder over a sliding window, which is S1. The Approach section calls what it encodes “attacker type”, which is S2. The type set exists only because you wrote Avoidant and Enticed, and the classifier head is supervised by a label the testbed hands over, which is S9.

“The opponent embedding τ encodes attacker type and recent attacker behavior, inferred only from defender-observable signals.”

main.tex, Approach, Preliminaries

“During training the testbed exposes ground-truth attacker class and the attacker’s action sequence as privileged signals.”

main.tex, Opponent Model

Three senses in one symbol, in one section. Reviewer C’s complaint that the opponent representation is not described consistently is the same observation arriving from outside.

D6 O9 against O11, the two types

First pass calls them one sense, second pass calls them two.

`next.md` says the cooperative and adversarial halves are “the same inference, recovering a hidden type from behaviour, while the other party adapts, done twice with different vocabulary”. That is your own statement that O9 and O11 are one thing.

The second pass, seeing the sentences without the file names, put O9 in S2 and O11 in S5, and the operational test agrees with the second pass: the categories on the Iterated Prisoner’s Dilemma page move when the group moves, and a hidden type in a Bayesian game does not. So the folding `next.md` proposes is not yet done. It is a research question stated as a fact, and the census cannot make it true.

D7 O13, the information structure

First pass S1, second pass not an occurrence.

“Nobody sees the strategy list, they only see what happened last round” describes what the agents can observe. The first pass read it as opponent modeling, because inference from local information alone is the defining property of S1 and of L2. The second pass read it as a property of the environment, which constrains any model but is not itself one.

The distinction matters for the census’s own boundary. If every statement of partial observability counts as an occurrence, half the FOE-Dreamer paper counts.

D8 O17, a population of what

First pass S9, second pass S3.

“train defenders against a population rather than a single idealised opponent.” Read from the Biased Attacker page, the population is a set of scripted profiles you wrote, which is S9. Read cold, a population of opponents that a policy trains against is the population in PSRO, and training against it is best response computation, which is S3.

The two passes put the same sentence in two of your own projects. AcceleratePSRO uses population in the second sense throughout. Nothing on the Biased Attacker page says which one it means, and the two carry different obligations: an S3 population needs a meta-strategy solver, an S9 population needs a coverage argument.

D9 O30 and O31, is not modeling a kind of modeling

First pass S3 and none, second pass none and S3, in that order.

O30 is “computing a best response inside PSRO”. O31 is “the opponent can be folded into the environment as one more stochastic process”. The first is optimization against a given opponent with nothing inferred; the second is the verdict that no opponent reasoning is needed at all.

The passes swapped. One of them treated best response as modeling because the opponent’s strategy is its argument; the other treated folding-in as modeling because deciding an opponent is stochastic is a claim about that opponent. Learn Structure’s opponent-blindness test is exactly this decision made explicit, which is why O32 lands in S3 in both passes while its two neighbours do not.

Your table’s Epistemic operation axis is what splits them. It runs describe, infer, predict, simulate/enact, and it has no position for optimize.

5.1 One split that is not a disagreement

O5 and O26 both land in S7 in both passes, and they still do different work. The dictionary row says threat modeling characterizes a threat space, often prospectively. Section 4 of the FOE-Dreamer paper uses its threat model to fix what the defender is not responsible for:

“Several threats are out of scope. We do not consider adaptive learning adversaries that retrain against the deployed defender; the two profiles are scripted and fixed-class.”

main.tex, Threat Model

Characterizing a space and bounding a claim are both S7, and the second is the one a reviewer reads. Worth knowing that your paper’s threat model is doing the bounding job, since the same paragraph is what Reviewer C used to ask for attackers that differ in other behaviors.

5.2 The constraint every sense shares

O33 and O26 say the same thing from two projects. Learn Structure:

“This property carries the section’s sharpest finding: none of the four is defined for an adapting adversary.”

taxonomy.tex, Properties of the Measures, Opponent

FOE-Dreamer puts adaptive learning adversaries out of scope in Section 4. So the fixed-opponent restriction is not a limitation of one paper. It is a boundary that both projects sit inside, and it applies to S1, S2, S3, S4 and S5 alike. S8 and S9 are the two senses it does not touch, because both build the opponent rather than infer it.

6 Five more rows

The table in Section 1 stands. What follows extends it, using the same five columns, with one row per sense that the original three do not cover. Read the two tables together.

Concept	What is represented?	Why represent it?	Relationship to reality	Output
Type inference (S2)	private information as one of a named set	choose play against who you actually face	posterior over a set fixed before play	a distribution, updated in play
Best response (S3)	a strategy already in the population	decide whether game-theoretic reasoning pays here	none, the opponent is given rather than estimated	a new strategy, returned to the pool
Behavioral adversary model (S4)	systematic departure from expected-utility play	predict the choices of a person rather than of a solver	parameters fitted to, or imposed on, human choice	a policy with psychological parameters
Attacker construction (S9)	a behavior you specify	supply a training or evaluation opponent, with labels	fidelity to a property, not to an actor	a running agent and its ground truth
Theory of mind (S6)	beliefs, goals, and beliefs about you	anticipate a reasoner who is anticipating you	attribute states nobody observes	a nested model, with a declared depth

6.1 What the geometry cannot show

Your dictionary page invokes conceptual spaces, and a concept there is a set of regions plus salience weights plus correlations between dimensions. The table records the regions. Two correlations it cannot record, both visible once the eight rows sit together.

Fidelity rises with Specificity, in seven rows out of eight. Abstract representations describe generic adversaries; executable reproductions reproduce particular actors. The exception is attacker construction. It is maximally executable and deliberately generic, a class rather than an actor, and that single off-diagonal cell is the reason it needs its own row instead of being read as a weak form of adversary emulation.

Epistemic operation has no position for optimize. The axis runs describe, infer, predict, simulate/enact. S3 does none of those. It takes an opponent as given and returns a policy, and D9 is what happens to a census when a sense has nowhere to sit. Either the axis gains a fifth position or S3 is declared out of the dictionary’s scope, and the second option costs you AcceleratePSRO.

7 Translation

The same object, in the three vocabularies. This is the table to hand a collaborator before the meeting rather than during it.

The object	Multi-agent RL	Game theory	Network security
the other agent's hidden characteristic	opponent embedding, latent	type	threat actor profile
the set of others you might face	opponent distribution, opponent population	type space	the actors in scope
the fixed other agent you train against	scripted opponent, environment agent	a strategy in the population	emulated adversary
your representation of them	opponent model	belief over types	threat model
information you have in training and not at test	privileged information, centralized training	complete information	ground truth from the range
checking the representation works	held-out opponents, unseen attackers	exploitability, regret	red team exercise
what breaks it	non-stationarity, an opponent that adapts	an opponent that best-responds to you	an actor that changes its techniques

Three fields, three outputs, and the outputs are the reliable tell. RL returns a vector. Game theory returns a distribution or a strategy. Security returns a document. When a conversation stalls, ask what the other person expects to hold in their hand at the end of the modeling, and the sense falls out.

Row four is where the trouble is, and it is worth saying plainly. Opponent model and threat model are not translations of each other. They are the same slot filled by different objects, one learned and one written, one consumed by a policy and one read by a person. Your table already says this in its Output column. The Headspace question is asking what the second can demand of the first, and that is a real question with no answer yet in any of the three fields.

8 What this file settles

Forty of forty-nine occurrences have one sense and keep it. Nine do not, and those nine are Section 5. Three of them, D2, D5 and D9, are places where the five dimensions themselves come up short: an Object with no position for algorithmic provenance, one symbol carrying three senses inside one section, and an Epistemic operation axis with no position for optimize.

The stub page names Threat Modeling and Adversary Emulation and stops. Both are here, S7 and S8, with the operational test that separates them and the seven other senses they were being separated from. The row your table never had is S9, and it is the one you use most.